

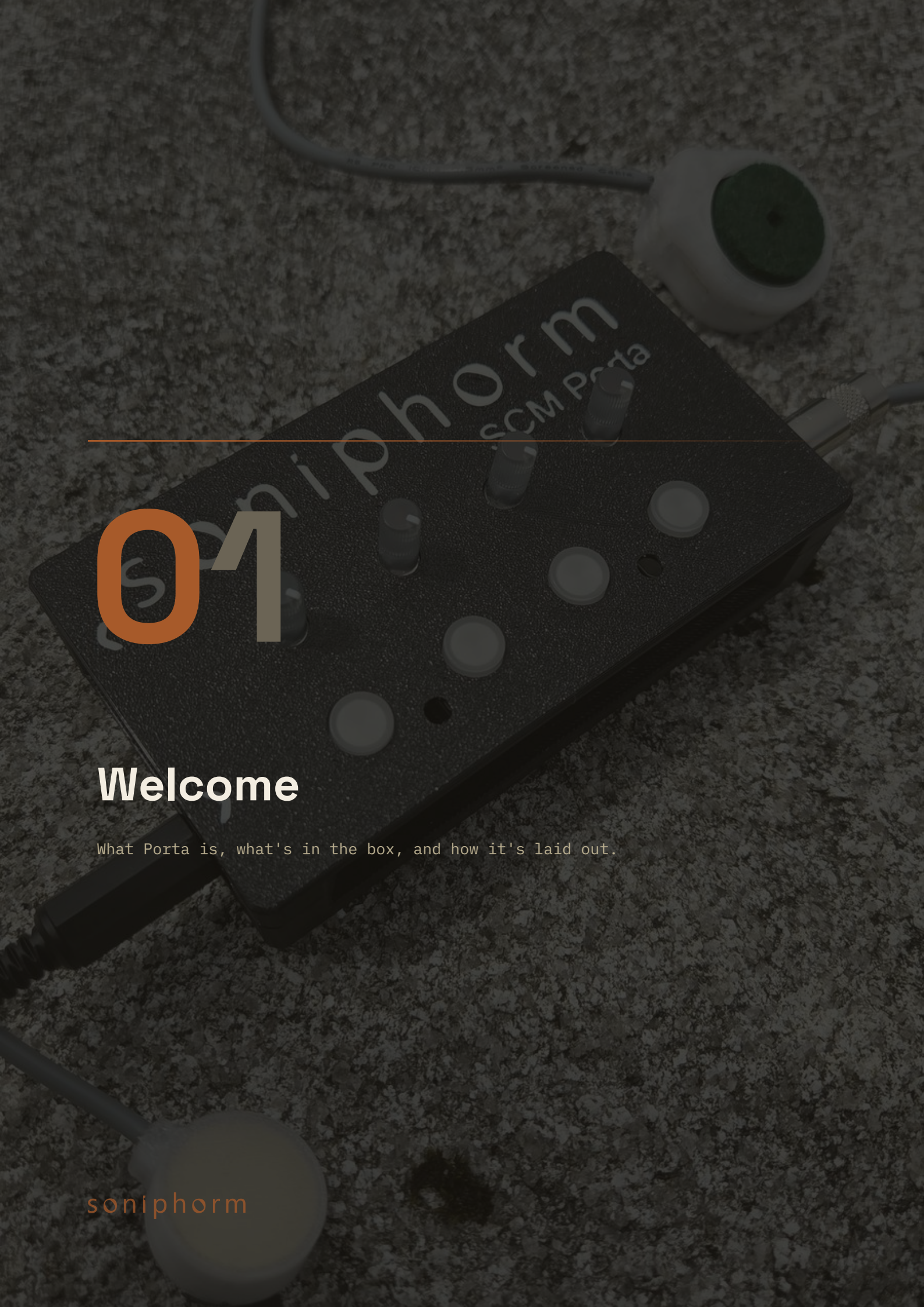
soniphorm

SCM Porta

OWNER'S MANUAL

Smart Contact Mic Porta: vibration sensing and processing on the move

Patch it visually in the Soniphorm Patcher, play it
by hand.

A black, rectangular electronic device with a textured surface, featuring several circular buttons and a small display. It is connected to a cable with a green circular connector. The device is resting on a dark, textured surface.

01

Welcome

What Porta is, what's in the box, and how it's laid out.

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What is a contact mic?

A contact mic senses vibration directly from whatever it's touching, rather than sound travelling through the air like a conventional microphone. For sound design, that distinction matters: you're capturing an object's raw mechanical vibration — a window, a radiator, a mop bucket — rather than the acoustic result of it after the room and the air have shaped it. It's a direct line into a surface's own resonance.

On their own, contact mics are simple and cheap to build — a piezo element is all it takes to sense vibration. But a bare piezo doesn't do much by itself: it needs the rest of a signal chain around it to be usable. Soniphorm's own Active Contact Mic, for instance, has its own impedance-matching preamp built in, but still needs external power and something to record or amplify into — a recorder, a mixer, an interface. Piezo elements also tend to misbehave with standard microphone preamps, especially at low frequencies, which is part of why contact mics have a reputation for sounding thin or harsh unless the signal is properly balanced first.

Porta folds all of that into one handheld unit. Piezo pickup, impedance matching, preamp, and a headphone-capable output are already built in — plug in a pair of headphones and you're listening to a properly balanced contact mic signal immediately, no mixer or interface required.

UNDER THE HOOD

The signal runs through an analogue front end into a digital stage on an ESP32-based microcontroller, sampling at 24-bit / 48kHz — or 96kHz, if you choose it. Battery powered and handheld throughout, with direct-to-SD recording built in.

Who it's for

Most of Soniphorm's contact mic customers so far have been sound designers working in film and TV, drawn to how directly it captures an object's own vibration. It also has an audience among performers who find pre-recorded or sampled material limiting on stage — Porta responds to whatever's actually in front of it, live, rather than triggering something captured earlier. More generally, it's a playground for anyone curious what ordinary objects sound like from the inside.

What is Porta

Porta is a Smart Contact Mic brain in a box: the hub that a separate piezo contact mic and a separate exciter plug into, with a real physical control panel on the front — four backlit RGB potentiometers, four backlit RGB buttons, and two bank-navigation buttons — plus hardware MIDI In and Out.

Porta itself never touches a surface. The piezo pickup and the exciter are each standalone units on their own ~70cm cables, terminated in balanced stereo TRS mini-jacks that plug into the box, rather than being built into a single handheld body.

Porta shares its core signal chain — piezo input, exciter output, patcher-driven DSP — with Soniphorm's DIY Smart Contact Mic kit. The difference is what you're holding: the DIY kit is the bare board plus a breakout of pots and buttons you wire up yourself; Porta is that same control surface built into one finished, battery-powered hub, with the piezo and exciter as dedicated cabled peripherals rather than a bare board and alligator clips.

HOW PATCHING AND PLAYING SPLIT

Patches — the actual DSP signal chain — are designed visually in the Soniphorm Patcher, then loaded onto Porta. Once a patch is loaded, the front panel takes over: its four pots and four buttons are what the patch exposes for live control, organised into banks you step through with the two navigation buttons.

Core capabilities

- **Piezo pickup + exciter, cabled** — separate units on ~70cm balanced TRS cables, so pickup and exciter can sit on the same surface or two different ones.
- **Physical control surface** — 4 backlit RGB pots (clear shaft) + 4 backlit RGB buttons + 2 bank-navigation buttons, live-controlling whatever the loaded patch exposes.
- **Swappable patches** — the signal chain is built visually in Soniphorm Patcher and loaded onto the unit.
- **PortaSampler** — up to 8 chromatic sample slots, playable and pitched from the patch.
- **Granular processing** — built-in granular engine for stretching, smearing, and re-texturing whatever the mic picks up.
- **On-board recording** — captures direct to SD card, no computer needed.

- **Hardware MIDI In/Out** — TRS Type A, for syncing or controlling external gear (or being controlled by it).
- **Battery powered** — plays untethered for a full session.

Patches load onto Porta over USB or BLE — see Connecting to Soniphorm Patcher, 2.2, for both paths.

01.3

In the box

- Porta hub (box, control panel, MIDI I/O)
- Piezo pickup — contact mic capsule on ~70cm balanced TRS mini-jack cable
- Exciter — standalone unit on ~70cm balanced TRS mini-jack cable

No USB cable is included — you'll need your own data-capable USB-C cable to charge and to patch/update firmware over USB. If you're connecting MIDI gear (3.5) and your other equipment uses standard 5-pin MIDI rather than TRS, you'll need your own Type A TRS-to-DIN adapters.



FIG 0.1 — Porta hub with piezo pickup and exciter.

Anatomy

Porta's shell is 3D-printed black PETG. The box is a hub: the piezo pickup and exciter connect to it as separate cabled peripherals rather than living inside it. Every feature is numbered below, read left side first, then clockwise around the box.

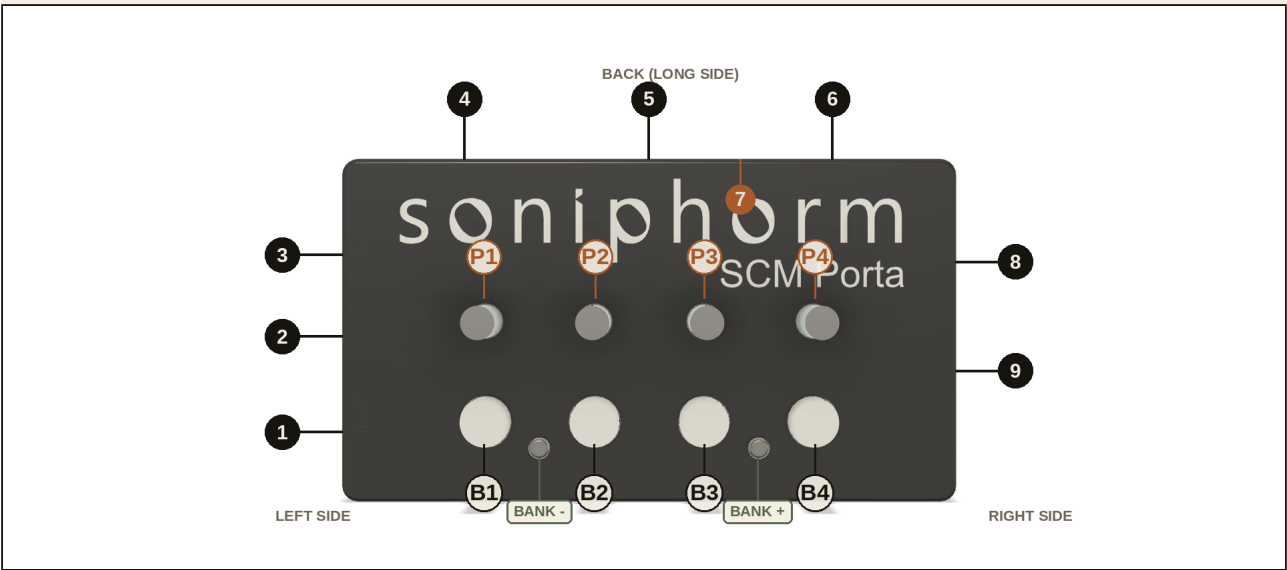


FIG 3.1 – Porta, full layout: jack positions (1-9) and control panel (P1-P4, B1-B4, bank nav), labelled directly on the actual render. Jacks 1-9 sit on the sides/back not visible from this top-down angle; callouts show which side each is on.

1 – Piezo in	TRS jack, balanced, for the piezo pickup's ~70cm cable. Etched "IN", under a shared "AUDIO" heading with item 3. A small white arrow points inward toward the jack.
2 – Boot / Reset	Two small buttons, etched "B" and "R", present on the board but not part of the normal firmware update process – see 6.3.
3 – Line / headphone out	Line-level output with enough drive to run headphones directly. Etched "OUT", under the same shared "AUDIO" heading as item 1. A small white arrow points outward from the jack.
4 – Power switch	Etched "I" and "O".

5 – MIDI In	3.5mm TRS, Type A wiring (see 3.5). Etched "IN", under a shared "MIDI" heading with item 6.
6 – MIDI Out	3.5mm TRS, Type A wiring (see 3.5). Etched "OUT", under the same shared "MIDI" heading as item 5.
7 – microSD slot	Sits between/under MIDI In and Out, etched "SD" underneath the slot. Recommended card: SanDisk Ultra microSD, 32GB, Class 10.
8 – USB	USB-C. Serial connection to the Soniphorm Patcher, and battery charging – one port, both jobs. Etched "USB" above the connector.
9 – Amplifier out	Feeds the provided exciter, etched "AMP". Wiring your own speaker: sleeve = amp–, ring = amp+, tip = not connected – matches Soniphorm's Active Exciter line.

NOTE

The B1–B4/P1–P4 labels used throughout this manual are our own reference labels for clarity – the buttons themselves aren't numbered on the panel.

Control panel

The top face carries two rows: 4 potentiometers (P1–P4, clear shaft, RGB backlit, the smaller caps) across the top, and 4 buttons (B1–B4, RGB backlit, the larger caps) across the bottom, each pot sitting above its corresponding button. Two small bank-navigation buttons are nestled just below the button row, between B1/B2 and B3/B4. These map directly onto whatever the loaded patch exposes – see Patches, pots & banks, 3.2, for how that link works.

- **4× potentiometers (P1–P4)** – continuous rotary controls. Each only lights up once turned above zero – at rest, an unused pot stays dark.
- **4× buttons (B1–B4)** – triggers, toggles, or momentary actions defined by the patch.
- **2× bank-navigation buttons** – step back/forward through 6 banks, paging P1–P4/B1–B4 through a much larger set of exposed parameters than 4-and-4 alone would allow.

The RGB backlighting indicates which bank is active:

Bank 1	Red
Bank 2	Green
Bank 3	Blue
Bank 4	Yellow
Bank 5	Cyan
Bank 6	Magenta

Some functions override this and keep a fixed colour regardless of bank — a button assigned to recording, for instance, stays red on any bank, since that's a widely understood colour for record.

TIP

On power-up, all the pots and buttons cycle through their lights once as a startup routine, confirming the unit has booted correctly.

Piezo pickup & exciter

Both are standalone units on their own ~70cm cables, terminated in balanced stereo TRS mini-jacks. Being separate from the box means the pickup and exciter can be placed on the same surface for direct feedback, or on two different surfaces — a tap picked up from a window and driven back into a radiator, for instance — limited only by cable reach.



FIG 3.7 — Piezo pickup.



FIG 3.8 — Exciter.

Piezo pickup

26mm × 8mm capsule, epoxy-sealed (safe to submerge in water). ~70cm cable to a 3.5mm

TRS jack plug with a plastic sheath. Housed in clear PETG.

Exciter

35mm diameter × 15mm height × 40mm length capsule. ~70cm cable to a REAN 3.5mm TRS jack plug. Housed in clear PETG.

The piezo pickup has a magnet built in, strong enough on its own for ferrous metal surfaces. For anything else, Blu Tack is usually enough — it's non-permanent and suits most surfaces, though tape or a small clamp can work better depending on what's being attached to.

The exciter is meant to be held against a surface by hand rather than mounted. These transducers are normally glued down permanently, but that would defeat the point here — Porta's exciter needs to move freely between surfaces. Its felt contact pad softens the mechanical vibration enough to make handheld use practical and comfortable, at some cost to raw output compared to a permanently fixed exciter.

DIY

Building or replacing your own piezo capsule (see Porta's shared signal chain with the DIY Smart Contact Mic kit, 1.1): wire **signal+ to tip**, the **brass disc to ring**, and the **shield of a 2-core shielded cable to sleeve** — the shield is not connected at the piezo end, only at the plug. For a replacement exciter/speaker instead: **amplifier+ to ring, amplifier- to sleeve** (tip not connected), matching the exciter output wiring in 06.2's spec table.

02

Getting Started

Charge it, power it, load a patch, hear it.

Charge & power on

1 Charge before first use.

Charge over USB-C (jack panel item 8). A charge-indicator LED sits at the bottom-right corner of the front panel, visible through a small light-pipe hole: red while charging, green once full. First charge takes about 2 hours.

2 Power on.

Flip the power switch (labelled I/O, back edge — jack panel item 4). All the pots and buttons cycle through their lights once as a startup routine, confirming the unit has booted correctly.

Connecting to Soniphorm Patcher

Patches are designed visually in the Soniphorm Patcher, built from the same module library documented at soniphorm.com/docs/modules, then loaded onto Porta over USB or BLE — either works for patching.

Over USB

- 1 Connect Porta to a computer via the USB port (jack panel item 8) using a data-capable USB cable (not included).
- 2 Open the Soniphorm Patcher at soniphorm.com/patcher, in a Chrome- or Edge-based browser (Web Serial support is required; Firefox and Safari won't connect).
- 3 Click **Connect**, top right of the patcher screen.
- 4 Select **Soniphorm Smart Contact Mic** from the list of USB devices.

Over BLE

- 1 Open the Soniphorm Patcher on a phone or tablet — Chrome on Android, or **Bluefy** on iOS (Safari doesn't support the Web Bluetooth connection the patcher needs, so Bluefy stands in for it).
- 2 Click **Connect** and select Porta from the Bluetooth device list.

NOTE

Firmware updates (6.3) are the one thing that still needs a computer over USB — but since it all runs in a browser with no install or login, it's a quick favour to ask if you normally patch over BLE.

TIP

The patcher canvas is dulled/dimmed when nothing is connected, and brightens once Porta is successfully paired — that's the visual cue to look for rather than a message or popup.

Once connected, a **Download Patch** button pulls whatever patch is currently loaded on the device back into the canvas — useful for picking up where a previous session left off, or checking what's actually on the unit. For the full step-by-step on building and uploading a patch once connected, see Building & uploading a patch, 3.2.

02.3

Your first sound

- 1 With a factory patch loaded, mount the piezo pickup to a resonant object — a table edge, a window, a mop bucket, anything with body — and the exciter to the same or a nearby surface.
- 2 Tap, scrape, or drag across the surface and listen for the exciter feeding back into it.
- 3 Turn the four pots and try the four buttons to hear what the patch has exposed — bank 1 first, then step through the remaining banks with the navigation buttons.

TIP

The clearest first sound usually comes from something with a bit of resonance and mass — glass, sheet metal, a hollow wooden box — rather than something soft or heavily damped.

03

Playing & Patching

Contact mic technique, patches, the panel, the sampler, and the granular engine.

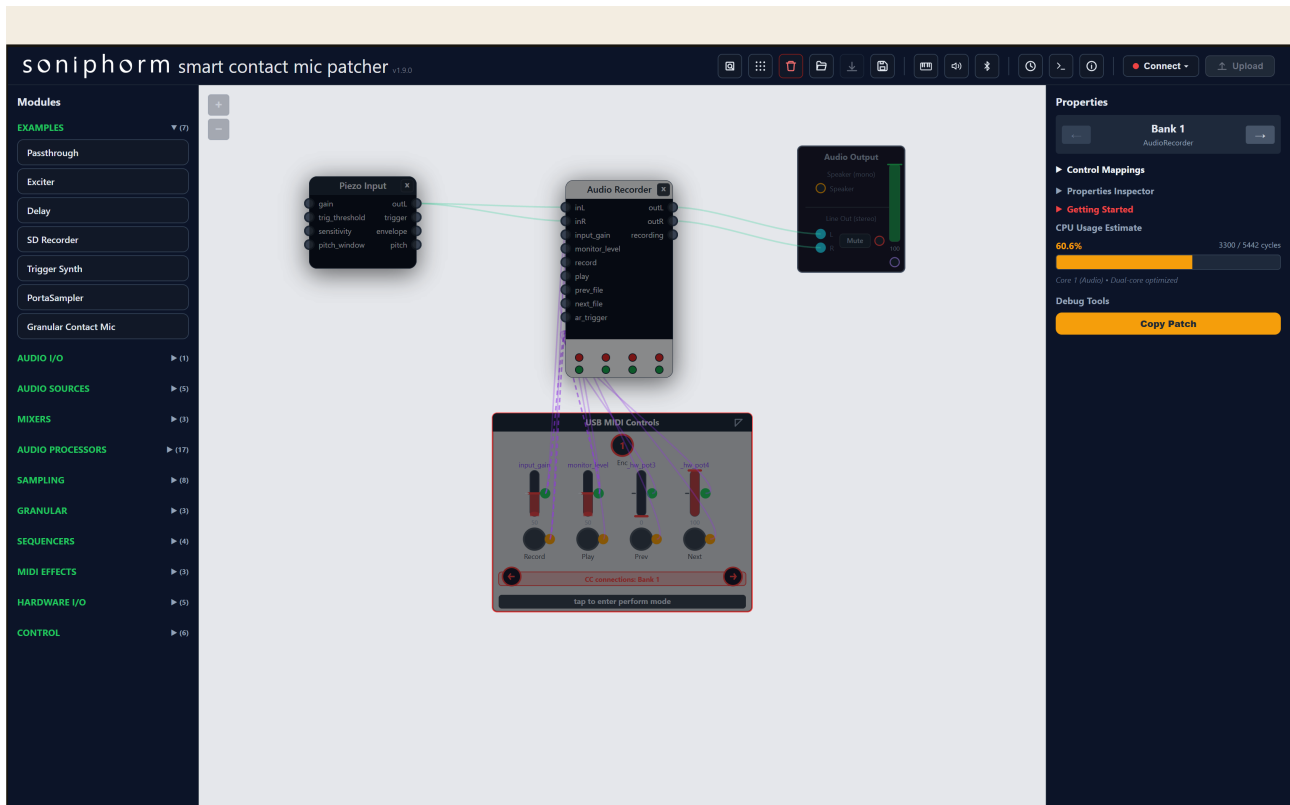


FIG 3.0 – Soniphorm Patcher, real screen capture: module index (left), canvas (centre), inspector (right), toolbar (top). Shown loaded with the built-in "SD Recorder" example patch – Piezo Input recording through Audio Recorder to Audio Output, with input_gain/monitor_level and the transport (Record/Play/Prev/Next) mapped onto the USB MIDI Controls strip.

03.1

Contact mic & exciter

The piezo pickup senses vibration directly through whatever surface it's mounted to, rather than sound through air – so isolation and contact/mounting quality matter more than distance or angle. The exciter drives audio back into whatever surface it's mounted to, turning that surface into a speaker. Because each is a separate cabled unit, they can be mounted to the same surface for direct feedback, or to two different surfaces entirely – cable length is the only limit.

Underneath, the piezo input does more than pass audio through: it also runs trigger detection, envelope following, and pitch tracking on whatever it hears, so a patch can respond to how a surface is struck, not just capture the sound of it – a hit above the trigger threshold can fire a sample or envelope, the envelope follower can shape a filter or volume in time with the dynamics, and the pitch tracker can follow a struck object's resonance into an oscillator. On

Porta, these are exactly the kinds of parameters you'd expect to find mapped to the front panel's pots and buttons.

- **Mounting quality** — however the pickup and exciter attach to a surface (see 3.6 below), a firm, consistent bond generally gives a stronger, cleaner signal than a loose or shifting one.
- **Surface choice** — hard, resonant materials (glass, metal, dense wood) carry more character than soft or heavily damped ones, and pitch-track more reliably.
- **Feedback loops** — with the exciter and pickup on the same or coupled surfaces, feedback is part of the instrument, not a fault. Repositioning either unit controls it — as does whatever gain pot the patch exposes.
- **Cable reach** — at ~70cm each, the pickup and exciter can be positioned well apart from Porta itself and from each other; plan the setup around what's actually reachable before starting.

CAUTION

At high exciter drive and close pickup/exciter coupling, feedback can build quickly and loudly. Start at low patch gain when exploring a new surface, especially through headphones.

03.2

Patches, pots & banks

A patch is the signal chain running on Porta — how the piezo input is processed before it reaches the exciter, the recorder, or both. Patches are built visually in Soniphorm Patcher from a shared library of 40+ modules (sound sources, effects, granular/looping, control, and I/O), then loaded onto the unit.

What makes a patch playable rather than just audible is how it maps its parameters onto the front panel. Certain modules are **auto-mapped**: drop one into a patch and it claims one or more hardware banks for itself, wiring its own parameters directly onto P1–P4/B1–B4 with no extra cabling required. PortaSampler, for instance, claims 3 consecutive banks; Freeze Ensemble claims 1. See the Module Reference (Appendix A) for the full auto-mapped list and exact pot/button assignments per module.

For anything else, there's **USB MIDI Controls**: patch a parameter to it and it becomes live on the panel, in whichever bank is currently free. It's not itself a module you pick from the palette — it's a fixed strip that's always present on the canvas, one of the most fundamental pieces of

the system since it's the direct link between a patch and the physical hardware. Under the hood each of its four slots is a MIDI Control module instance (the same one listed in A.4, one CC per instance, `midi_source: usb`), which is why the module reference shows a single out port with no inputs — you don't patch into MIDI Control directly, the strip creates and wires the instances for you. It's unrelated to the MIDI In / MIDI Out modules, which are the hardware TRS jacks rather than this panel-mapping mechanism.

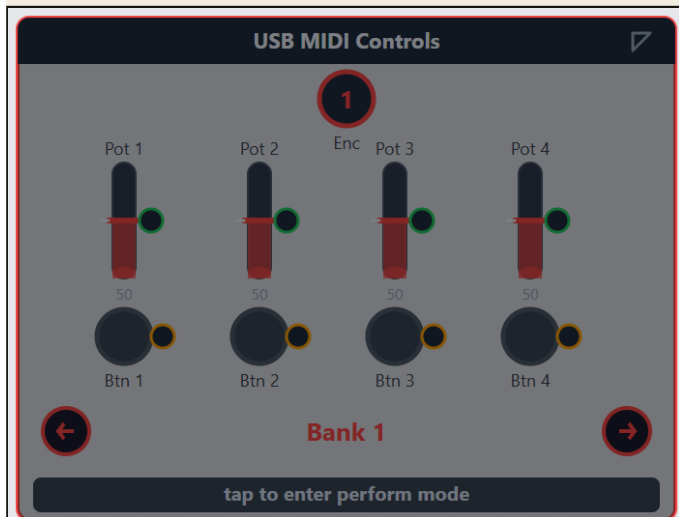


FIG 3.2a – USB MIDI Controls strip, real screen capture. Four pot mappings (here labelled Kick/ Snare/Hat/Perc by an auto-mapped Drum Machine), the bank encoder, bank-navigation arrows, and the current bank readout.

P1–P4 and B1–B4 (see 1.4) are organised across **6 banks**, stepped through with the two bank-navigation buttons. A simple patch might live entirely in bank 1; a deeper one can spread gain, tone, sampler, and granular controls across all 6.

Building & uploading a patch

- 1 Connect to Porta over USB or BLE (see 2.2) — check the panel lights up, confirming the connection is live.
- 2 Either open one of the built-in patches from the Examples list in the module library, or build your own: add modules to the canvas and connect them — green cables carry audio, purple cables carry data/control.
- 3 Patch any parameters you want live on the panel to the USB MIDI Controls module — modules that auto-map (3.3, 3.4) don't need this step.
- 4 Click **Upload**, top right of the patcher screen.

- 5 Name the patch when prompted — worth doing properly so you can find it again later.
- 6 Upload takes a few seconds; once it completes, Porta is running that patch.

CAUTION

Porta's processor has a limited CPU/PSRAM budget, and it's easy to exceed it by stacking multiple demanding modules in one patch — two or more reverbs, for instance. The patcher's inspector shows a live CPU usage estimate for the current patch, and flags a warning when a patch is over budget. If the processor can't keep up, back off the module count or swap a lighter alternative in rather than uploading anyway.

Porta stores up to **4 patches on-device**, in hardware slots 1–4. The upload dialog lets you name the patch and choose which slot to save it into — slot 1 is preselected and is what every upload overwrites by default if you don't touch the slot picker, so a single-patch workflow needs no extra steps.

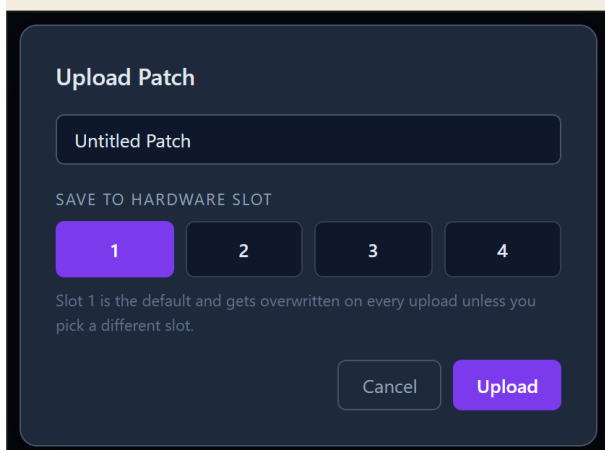


FIG 3.2b — Upload dialog: name the patch, pick a hardware slot (1–4). The slot picker only appears once Porta's firmware is new enough to support it (1.9.52+).

To switch between saved patches from the panel itself: tap a bank-navigation button repeatedly to cycle one position past your last real bank — Porta enters patch-select mode automatically. Btn1–4 then show the saved slots — solid green for the slot currently loaded, solid red for another occupied slot, unlit for empty (an empty slot can't be selected). Press a red (other-saved) button to load that patch and return to normal play; cycling the bank button back down leaves patch-select mode without loading anything.

03.3

PortaSampler

PortaSampler is an 8-slot sample recorder and player. Each slot records from the piezo input using threshold detection with pre-roll transient capture — arm a slot, make a sound, and recording starts automatically when the input crosses the threshold and stops when it dies away, so the transient at the start of the hit is never clipped off.

- **One sample per slot** — the normal way to use PortaSampler: 8 independent slots, each holding one recording.
- **Three playback modes** — one-shot, loop, and velocity-layer (velocity-layer is an advanced/kit-file feature — up to 3 recorded takes per slot, selected by how hard the slot is triggered, MIDI velocity included — and isn't exposed on the panel).
- **Kits** — a full set of 8 slots can be saved to and loaded from the SD card as a kit, so you can switch between sample sets without re-recording.

PortaSampler claims 3 consecutive hardware banks:

Bank 1 — Record

P1 = threshold, P2 = pitch, P3 = mode (one-shot/loop), P4 = volume. B1 = arm/trigger/stop (short press) · clear (hold 500ms+ — wipes the recording and leaves the slot idle; press B1 again to re-arm, same as any other empty slot), B2 = reverse, B3 = previous slot, B4 = next slot.

Bank 2 — Trim

P1 = start, P2 = end (relative to Start, not the absolute sample — 0 sits right at Start, full winds out to the end of the sample, so the pot's full travel stays useful no matter where Start is set), P3 = pan, P4 = aux send. B1 = trigger the selected slot, B2 = mute the selected slot, B3/B4 = previous/next slot.

Bank 3 — Filter / Trigger

P1 = destructive filter cutoff, P2 = filter type (HP/BP/LP). P3-P4 unassigned. B1 = trigger the selected slot, B2 = apply the filter to the selected slot, B3/B4 = previous/next slot.

B1 (trigger) and B3/B4 (previous/next slot) work identically on all three banks — only B2 differs (reverse on Bank 1, mute on Bank 2, apply filter on Bank 3), and every button always acts on whichever slot is currently selected, not a fixed slot number.

- ① On Bank 1, set the threshold with P1 and press B1 to arm.
- ② Make a sound against the surface the piezo pickup is mounted to — recording starts and stops automatically around the transient (up to the recording's maximum length, set from Soniphorm Patcher's Inspector panel — no dedicated pot).
- ③ Navigate slots with B3 / B4 (the same on every bank). Trigger the selected slot with B1 on Bank 2 or Bank 3, mute it with Bank 2's B2, or destructively apply the filter with Bank 3's B2.

Screen-free feedback, on Bank 1: B1 flashes red whenever the selected slot is empty and ready to capture — armed or not, freshly cleared or freshly navigated to, it's the same unambiguous cue — and goes solid red while actually recording. B2 shows reverse as a held state, not a flash: solid red means the selected slot is set to play backwards. P1 doubles as a live input meter while a slot is empty (green above the record threshold, red below, so the threshold can be dialled in and the target sound previewed by ear and eye before committing) and switches to slot status once it holds a take (solid blue = has a recording, green = that slot is currently playing). Pressing B3/B4 to change slots also lights up every Bank 1 pot and button for one second to show exactly which of the 8 slots is now selected, then returns to normal.

Loop mode: triggering a looping slot again while it's already playing stops it rather than layering a second copy on top — press to start, press again to stop, from any bank's B1 (Bank 1's arm/trigger/stop button, or Bank 2/3's trigger button — they all trigger the same way). Playing a loop-mode slot from MIDI instead follows the note itself as a gate: note-on starts it, note-off stops it, matching how the CV gate inputs already behave. One-shot and velocity-layer slots aren't affected by any of this — retriggering those always plays another voice.

TIP

All 8 slots respond to MIDI notes 36–43 straight away, with real velocity driving playback level — no patch cable required, no MIDI module needs adding to the patch at all.

Elektron-style GM-drum note numbering: slot 1 = note 36, slot 2 = 37, and so on up to slot 8 = note 43. On a slot in velocity-layer mode, that same incoming velocity also picks which recorded take plays (soft/medium/hard maps to the 2nd or 3rd recorded layer, same selection the physical trigger uses) — no separate wiring needed for MIDI to reach the layers.

MIDI CC automation, per slot: beyond the panel, each of the 8 slots has 5 dedicated CC numbers — pitch, pan, volume, start, and end — that control that specific slot directly, no

matter which bank the panel happens to be showing. This is separate from the panel's own CC range (20–48, only live while its bank is on screen): these 40 CCs are always active, so an external sequencer can automate one slot continuously while the panel is being used for something else, with no conflict between the two. A slot's normal panel-set value is left alone until its automation CC has actually been received at least once, so turning this on can't silently reset a slot you aren't automating.

CC layout

```
CC = 50 + (slot-1)×5 + offset. Offsets: pitch=0,  
pan=1, volume=2, start=3, end=4. Slot 1 = CC50–54,  
slot 2 = CC55–59, ... slot 8 = CC85–89.
```

TIP

This pairs well with per-trig parameter locks on a step sequencer like an Elektron Digitakt: dedicate a CC lane per parameter on the track already firing the 8 note-triggers, and each trig can carry its own pitch/pan/volume/start/end alongside its note — turning a short one-shot into something closer to a chromatically-playable, loop-scrubbable instrument, sequenced entirely from outside Porta.

Kit browsing from the panel: beyond saving/loading kits over serial (PSSAVE / PSLLOAD), the 7 most recently saved kits can be browsed and loaded from the panel alone, no patcher connection needed. On Bank 1 (Record), hold both bank-navigation buttons together for 1 second to enter kit-select — the 8 Bank 1 pot/button LEDs repurpose to show 8 positions: Blank (wipes to an empty kit) plus the 7 most recent kits by save time, magenta on the pots and red on the buttons, with only the currently-highlighted position lit (fast-flashing if it's Blank or an empty position, solid if it's a real kit). Use B3/B4 to move between positions. Confirm and load the highlighted kit either by holding both bank-navigation buttons together again, or simply by leaving it alone for a couple of seconds — it auto-confirms once idle, so browsing doesn't require a second deliberate gesture to commit.

If the Inspector's `preview_enabled` option is on, each occupied kit auto-plays its slots in sequence while you browse past it — one-shots play once then move to the next occupied slot, loops play for roughly 2 seconds then advance — so a kit can be auditioned by ear before committing, without disturbing whatever kit is actually loaded. Off by default.

03.4

Granular processing

Porta's granular engine breaks incoming audio — live from the contact mic or from a PortaSampler slot — into small grains (roughly 10–100ms each) that can be stretched, re-pitched, and re-triggered. Built around the platform's Freeze Ensemble module, four

independent freeze layers share a stereo capture buffer, so up to four overlapping textures can be frozen at once.

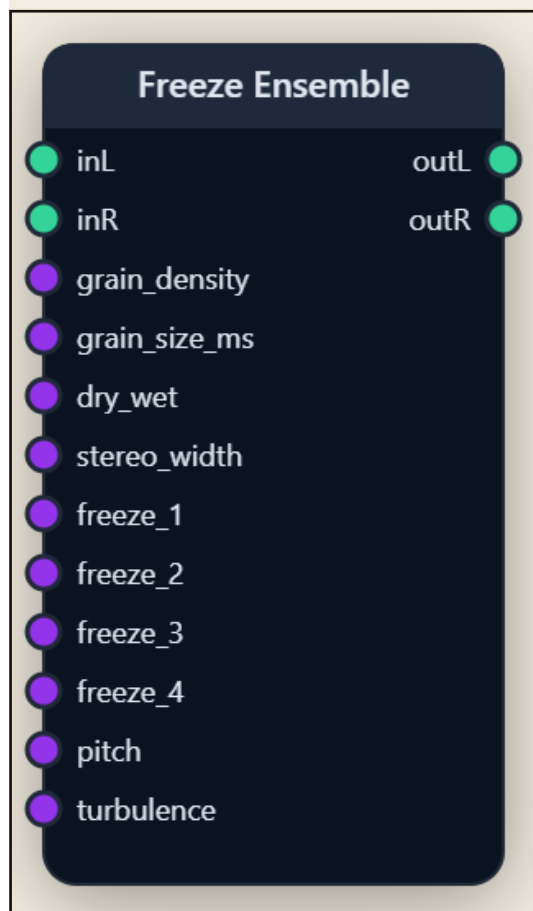


FIG A.1 – Freeze Ensemble, actual module.

The module's full parameter set is larger than the 4 pots can expose directly:

- **Grain density & size** — controls the rate and length of grains; shorter grains give sharper, more granular-sounding frozen textures.
- **Freeze 1–4** — latches each layer as a memory snapshot independently; trigger again to unfreeze.
- **Pitch** — shifts a frozen layer's playback, roughly ± 2 octaves.
- **Dry/wet** — balances the frozen/granulated signal against the dry input.
- **Stereo width** — spreads the frozen layers across the stereo field.
- **Turbulence** — randomises grain playback for a more restless, evolving texture.

Freeze Ensemble claims a single hardware bank for the four most performance-relevant controls: P1 = grain density, P2 = grain size (ms), P3 = pitch, P4 = dry/wet. B1–B4 freeze/

unfreeze layers 1–4 individually. Stereo width and turbulence aren't on the panel by default — patch them to USB MIDI Controls (3.2) if you want them live too.

TIP

Freezing a tap mid-decay and slowly turning up P2 (grain size) or P3 (pitch) is a quick way to turn a single strike into an evolving pad.

03.5

Connecting MIDI gear

Porta's MIDI In and Out let it sync with or control external instruments and drum machines, or be played/clocked from an external MIDI source. Both ports are wired to the Type A TRS standard (see 3.5's connector notes) — connect directly to other Type A TRS-MIDI gear with a plain stereo cable, or via a Type A TRS-to-DIN adapter for standard 5-pin MIDI equipment.

CAUTION

Don't assume a device's TRS MIDI port matches Porta's wiring. Type A and Type B TRS MIDI use opposite tip/ring assignments and are not interchangeable without a converter — check your other device's documentation before connecting.

Notes

Both directions, but neither is automatic — each requires a MIDIIn/MIDIOut (or USB MIDI In/Out) module explicitly wired into the patch. Receive: an incoming Note On/Off updates the module's note/gate/velocity outputs, for playing the synth from an external keyboard or sequencer. Send: whatever gate/note signal a patch wires into MIDIOut's inputs goes out as real Note On/Off messages — Porta doesn't send notes on its own initiative.

CC from the panel pots

Internal by default, not transmitted. Turning a panel pot writes its value into the same internal 0–127 CC slot that an incoming external CC message would use — one code path handles both, which is why USB MIDI Controls can equally be "played" by a physical pot or an external controller. But that's an

	internal representation only: a pot movement does not automatically go out over MIDI to a connected computer/DAW. To actually transmit a pot's movement out, a patch has to explicitly route it through a MIDIOut module's CC input – deliberate patch wiring, not automatic.
CC from external gear	Yes, and this is the primary intended direction – MIDIControl reads incoming CC (or note-gate) from either hardware TRS or USB MIDI and outputs it as a 0–1 control signal into the patch. This is how the buttonless Smart Contact Mic gets "virtual knobs" from a DAW or controller; on Porta it's an alternative or overlay to the physical panel, using the same CC numbers (20–43 across banks 1–6, base $20 + (\text{bank}-1) \times 4$) as USB MIDI Controls. PortaSampler additionally has its own always-on range, CC50–89, for bank-independent per-slot automation (pitch/pan/volume/start/end $\times 8$ slots) – see 3.3.
Clock	Both directions, over both hardware TRS and USB simultaneously – real 24-PPQ MIDI clock (0xF8) plus Start (0xFA) and Stop (0xFC) transport messages, not just tempo. Porta's Clock module can drive external gear or sync to an external clock source, either port.
Program Change	Received at the protocol level (correctly parsed so it doesn't desync the running MIDI stream) but not acted on anywhere – no patch/bank switching is wired to it. Porta never sends Program Change either. Patch switching between the 4 on-device slots (3.2) is a physical panel gesture only (cycle a bank-navigation button repeatedly past the last real bank), not a MIDI message.

04

Recording

Capturing straight to SD card.

SD card & recording

Porta records its output directly to SD card, with no computer or phone needed to capture a session. Recordings are saved as WAV files with an auto-incrementing filename, so nothing gets overwritten between takes.

- **Card format** — FAT32. Cards formatted exFAT or NTFS won't be recognised.
- **Card size** — 32GB or smaller is recommended for best compatibility. Recommended card: SanDisk Ultra microSD, 32GB, Class 10.
- **File naming** — each take is written as its own WAV file with an incrementing index, so recordings stack up rather than overwrite.

The card slot is on the back edge (jack panel item 7), between MIDI In and Out.

Recording and playback both run through the Audio Recorder module, which auto-maps a bank: B1 = record, B2 = play, B3 = previous file, B4 = next file. The same module handles browsing and playing back files already on the card, not just capturing new ones.

Record time: Audio Recorder always writes stereo, 24-bit PCM WAV (uncompressed), so time per GB depends only on sample rate:

48kHz (default)	~58 minutes per GB · a 32GB card holds roughly 31 hours
96kHz (optional)	~29 minutes per GB · a 32GB card holds roughly 15-16 hours

TIP

Format new cards as FAT32 before first use rather than assuming out-of-the-box formatting — most cards over a few GB now ship exFAT by default.

05

Care & Maintenance

Battery, cleaning, and what not to do to the enclosure.

Battery & charging

Charge over USB-C (jack panel item 8). The same LED that shows charge state also serves as the battery indicator, at the bottom-right of the front panel, visible through a small light-pipe hole: red while charging, green once full. First charge takes about 2 hours.

Battery	Lithium-ion, 1800mAh
Play time	~3–4 hours per charge
First charge	~2 hours

There's no distinct low-battery warning during play — no separate LED state or on-screen indicator beyond the charge LED's red/green charging status, and no low-battery detection at all. Porta will simply run until it powers off; there's no advance notice.

CAUTION

Standard lithium battery precautions apply — don't puncture, crush, expose to heat above [confirm rating], or leave on charge unattended for extended periods.

Cleaning & handling

- Wipe the enclosure with a dry or lightly damp cloth. Avoid solvents — the printed PETG shell can be attacked by some cleaners.
- Avoid fitting Porta inside a metal case or pouch with metallic lining, or adding metal hardware to the shell — the enclosure is deliberately non-metallic so it doesn't interfere with the BLE antenna.
- Keep the panel and connector faces dry — Porta is not rated for water exposure.
- Store away from prolonged direct heat or sunlight, which can soften PETG over time.
- Turn pots by the clear shaft rather than levering at the base, and avoid pressing buttons with anything sharper than a fingertip.
- Unplug the piezo pickup and exciter by the connector body, not by pulling the cable — a 70cm run has enough leverage to stress the jack over time.
- Avoid sharp bends or coiling the piezo/exciter cables too tightly when packing them away.

IP rating: Porta has no official IP/ingress rating — nothing is certified or tested against water or dust. Keep the Porta unit and the exciter dry; neither is water-safe. The **piezo pickup is the one exception:** its capsule is epoxy-sealed inside its clear PETG housing, genuinely safe to submerge even without a formal IP number.

06

Reference

Troubleshooting, specifications, safety, and support.

Troubleshooting

Symptom	Try this
Soniphorm Patcher canvas looks dim/washed out	Porta isn't connected. Click Connect (top right) and select Porta from the device list (USB or BLE) — the canvas brightens once paired.
Porta doesn't appear in the USB device list	Confirm the USB cable is data-capable, not charge-only. Use a Chrome- or Edge-based browser — Firefox and Safari don't support Web Serial.
Porta doesn't appear over BLE on iPhone/iPad	Safari doesn't support the Web Bluetooth connection the patcher needs — use the Bluefy browser on iOS instead.
Firmware update option isn't available	Firmware updates require a USB connection specifically — BLE won't offer it. Connect via USB to update, then switch back to BLE for everyday patching if you prefer.
No sound from the exciter	Check a patch is actually loaded, and gain within the patch — check the relevant pot across each bank — isn't at zero. Confirm contact between exciter and surface.
Weak or noisy piezo signal	Check the pickup's mounting quality and surface choice (see 3.1). Confirm the mounting surface is clean and dry, and the cable connector is fully seated.
Pots or buttons seem to do nothing	Confirm you're on the bank the patch actually maps them in — step through all banks with the navigation buttons.
MIDI gear not responding	Confirm both devices use the same TRS MIDI wiring standard (Type A vs Type B — see 3.5). A Type A device plugged into a Type B port (or vice versa) will not communicate without a converter cable.
SD card not detected	Format the card as FAT32 (not exFAT or NTFS). Use a card 32GB or smaller. Re-seat the card.
Clicking, glitching, or dropouts in audio	A heavily loaded patch can push the processor hard enough to glitch — try a simpler patch or lighter granular/reverb settings first.

Recording doesn't start / no file on card	In order of likelihood: (1) SD card wasn't detected at boot — it's only mounted on startup, so a card inserted after power-on won't work until Porta is restarted; (2) the Audio Recorder module is in threshold-triggered mode rather than manual, which deliberately disables the B1 record button — check the patch; (3) the card is full or write-locked, which silently rolls the recording back to idle rather than erroring visibly.
Sound cuts out or seems to disappear	Check the battery — a low charge can cause the audio to cut out. Put Porta on charge for a while; it's never a bad first thing to try.



06.2

Specifications

Enclosure	3D-printed PETG, black
Control panel	4× RGB backlit potentiometers (clear shaft), 4× RGB backlit buttons, 2× bank-navigation buttons, 6 banks
Piezo input	TRS, balanced, ~70cm cabled pickup
Exciter output	Amplifier out, ~70cm cabled exciter (sleeve = amp-, ring = amp+, tip = n/c)
Line/headphone out	Line-level, headphone-capable
MIDI	Hardware MIDI In + Out, 3.5mm TRS Type A
Patching connection	USB (serial, Chrome/Edge) or BLE (Chrome on Android, Bluefy on iOS)
Firmware updates	USB only
Recording	Direct to microSD, WAV, FAT32, ≤32GB. Recommended: SanDisk Ultra microSD, 32GB, Class 10
Sample slots	Up to 8, chromatic (PortaSampler)

Processing	Swappable patches, granular engine
Power	Li-ion, 1800mAh, ~3-4hr play time. Charge over USB-C (~2hr first charge), patch over the same port.
Dimensions	112 × 63 × 40mm (L × W × H), base unit only – excludes piezo pickup and exciter cables/units
Weight	70g, base unit only – excludes piezo pickup and exciter cables/units
MCU	ESP32
Sampling	24-bit, 48kHz (96kHz optional)
Price (at publication)	€250

06.3

Firmware updates

Firmware updates need Porta connected over USB specifically (not BLE) — everything else about connecting still happens entirely through Soniphorm Patcher in a browser, no software install or login, so it's a quick one-time trip to a computer even for someone who normally patches from a phone.

- 1 Connect Porta to Soniphorm Patcher over USB (see 2.2). If the connected unit is running outdated firmware, a notification appears via the **Device Info** button.
- 2 Click **Device Info**, then **Update Firmware**, then **Flash Firmware**.
- 3 Disconnect the USB cable, then reconnect it.
- 4 A new port appears in the browser's connection picker, labelled **USB JTAG** — select that one.
- 5 Once the flash completes, power the unit off and back on via the I/O switch (jack panel item 4).

NOTE

The Boot and Reset buttons (jack panel item 2) are present on the board but aren't part of this process — the whole update runs through the patcher and a USB reconnect/power cycle.

06.4

Safety notes

- Contains a lithium battery — do not puncture, crush, incinerate, or expose to excessive heat.
- Not rated for water or dust ingress — keep away from liquids.
- Keep away from young children — small parts and a battery are present.
- Exciter output can be surprisingly loud when driving a resonant surface directly against the body (e.g. held to an ear) — treat it with the same caution as headphones at volume.

06.5

Support & community

The discord.gg/yAFWg4eQXQ Discord is the main line for support — patches, troubleshooting, warranty claims, and sharing what you're building with Porta. It's a small, hands-on operation, so support comes directly from the person who built the unit, not a ticket queue.

Returns: 28 days from receipt.

Warranty: if Porta develops a fault due to how it was built — not damage from a drop, liquid, or modification — it'll be repaired or replaced, no argument. Reach out on Discord with what's happening and it'll be sorted from there.



Module Reference

The patcher's module library, and how it maps onto Porta's panel.

The module library

Every patch on Porta is built from Soniphorm's shared module library — the same one used across the whole patcher platform, documented in full at soniphorm.com/docs/modules. This appendix summarises it and highlights the modules that matter most for how Porta actually plays; it isn't a replacement for the live docs, which cover every module's full parameter set and get updated as the library grows.

Categories

Sound sources	Oscillators, noise, and input modules — including Piezo Input, the module carrying Porta's contact mic signal into a patch.
Effects	Filters, dynamics, and time-based processing applied to a signal in the chain.
Granular / looping	Grain-based and buffer-based processors, including Freeze Ensemble — the module behind Porta's granular engine (see 3.4).
Control	Envelope followers, triggers, sequencing, and control-rate logic — including PortaSampler (see 3.3) and USB MIDI Controls (see below).
Hardware I/O	Modules that talk to Porta's physical world: audio recording to SD, MIDI I/O, and the piezo/exciter jacks themselves.
MIDI effects	Processing applied to MIDI data passing through a patch.



Auto-mapped modules & the panel

Most modules just process signal inside a patch. A smaller set are **auto-mapped**: place one in a patch and it claims one or more of Porta's 6 hardware banks automatically, wiring its own

parameters onto P1–P4/B1–B4 with no extra patching required. Everything else reaches the panel by being patched to the USB MIDI Controls module instead (see 3.2) — auto-mapping just means a module skips that step. In the patcher itself, a red "Bank N" footer on a module block is the visual tell that it's auto-mapped (A.4).

PortaSampler	Claims 3 banks (Record / Trim / Filter-Trigger). Full mapping in 3.3.
Freeze Ensemble	Claims 1 bank (density, size, pitch, dry/wet + 4 freeze buttons). Full mapping in 3.4.
Audio Recorder	Claims 1 bank: B1 = record, B2 = play, B3 = previous file, B4 = next file. Full mapping in 4.1.
BPM Looper	Claims 2 banks. Bank 1: P1–4 = slot 1–4 volume, B1–4 = per-slot record/play/mute (tap cycles state, hold 1.5s clears). Bank 2: P1–4 = slot 1–4 slice-randomization amount, buttons unused (pitch is CV-only, no panel control).
Clocks	Claims 1 bank (2 if use_outputs_5_8 is on). Bank 1: P1–4 = output 1–4 clock division ($\div 16$ to $\times 16$), B1 = start/stop, B2 = tap tempo, B3 = reset to beat 1, B4 unused. Bank 2 (conditional): P1 = divisions for outputs 5–8, rest unused.
Drum Machine	Claims 2 banks. Bank 1 – Trigger/Pitch: P1–4 = kick/snare/hat/perc pitch, B1–4 = trigger kick/snare/hat/perc. Bank 2 – Volume/Mute: P1–4 = kick/snare/hat/perc volume, B1–4 = mute kick/snare/hat/perc.
Arpeggiator	Not auto-mapped. It isn't in the patcher's auto-bank list at all, so it reaches the panel the same way as any non-auto-mapped module: patch it to the USB MIDI Controls module (3.2).
Audio Slicer	Claims 3 banks. Bank 1 – Main: P1 = pitch (± 1 octave, centre detent), P2 = volume (centre = unity), P3–4 = start/end (reserved, not yet wired up), B1 = record/play (short press) or clear/load (long press), B2 = forward/reverse

	(short) or save to SD (long), B3/B4 = previous/next slice. Bank 2 – Slice ADSR: P1-4 = attack/decay/sustain/release, same B1-4 as Bank 1. Bank 3 – Aux Send: P1 = aux send level, P2-4 unused, same B1-4 as Bank 1.
Gene Looper	Claims 1 bank. P1 = gene selection (scrubs through divisions), P2 = speed/pitch (centre = unity), P3 = diffuse (per-slot allpass blur), P4 = pan. B1 = start/stop recording the current slot, B2 = cycle gene count (2/4/8/16/32/64), B3 = reverse toggle, B4 = advance to the next of 4 recording slots.
Harmonic Sequencer	Claims 2 banks. Bank 1: P1-4 = step 1-4 pitch (2-octave range each), B1-4 = toggle mute for step 1-4. Bank 2 is reserved but has no documented pot/button mapping.
Matrix Mixer 2x2	Claims 1 bank. P1-4 = the four routing amounts (A→A, A→B, B→A, B→B), bipolar -1 to +1 each. Buttons unused.
MIDI Looper	Claims 1 bank. P1 = playback velocity, P2 = transpose (±24 semitones), P3 = timing randomization, P4 = pitch randomization. B1 = record/play state machine (arm → count-in → record → play → overdub → stopped), B2 = hold for scale-quantize lock, B3 = long-press to clear, B4 unused.
Modal Resonator	Not auto-mapped. It has hardware-mapping code but is deliberately excluded from the auto-bank list (manual assign only), unlike every other module on this page. Reaches the panel via USB MIDI Controls (3.2), same as Arpeggiator above.
Multi Sample	Claims 1 bank. P1-3 = voice 1 / voice 2 / voices 3-4 levels, P4 = sample bank select (0-3). B1-4 = trigger voice 1-4.

Arpeggiator and Modal Resonator carry a "Bank 1" tag in the patcher UI but aren't actually auto-mapped — check the patcher's own auto-bank list rather than that tag alone if this ever needs re-verifying.

A.3

Modules behind Porta's core features

Piezo Input	Carries the piezo pickup's signal into a patch. Also runs trigger detection, envelope following, and pitch tracking on the incoming signal – see 3.1.
Audio Recorder	Writes output to SD as WAV, auto-incrementing filenames per take – see 4.1.
PortaSampler	8-slot sample recorder/player, one sample per slot, threshold + pre-roll capture, per-slot MIDI CC automation, kit save/load – see 3.3.
Freeze Ensemble	4-layer granular freeze engine behind Porta's granular processing – see 3.4.
USB MIDI Controls	The manual-mapping bridge to the panel – patch a parameter to it and it's live on P1-P4/B1-B4. Also reads incoming MIDI CC/notes, so it doubles as how Porta is played from an external controller – see 3.2.

For the complete, current module list with full parameter documentation, see soniphorm.com/docs/modules.

A.4

Module reference, by category

Every module, grouped the same way as the live docs at soniphorm.com/docs/modules, with a full port breakdown pulled directly from the module registry. The copper "Auto-mapped" badge means the module claims a hardware bank automatically (A.2) rather than needing to be patched to USB MIDI Controls.

IN PROGRESS

Stereo Out has no screenshot yet, since it isn't addable from the module palette (see its card for why).

PortaSampler

AUTO-MAPPED • 3 BANKS

8-slot sample recorder/player, one sample per slot. Records via threshold detection with pre-roll transient capture; plays back from gates, MIDI notes, or the panel. One-shot/loop/velocity-layer modes, kits saved/loaded from SD, per-slot MIDI CC automation independent of the panel banks.

INPUTS

- audioIn** recording source
- midi_in** optional channel scope
- gate1-8** slot triggers (5-8 hidden by default)
- pitch_cv** pitch offset, all voices

OUTPUTS

- mixL / mixR** main mix
- auxL / auxR** aux send



Bank 1 – Record	P1 threshold · P2 pitch · P3 mode · P4 volume · B1 arm/trigger/stop (hold=clear, then B1 to re-arm) · B2 reverse · B3/B4 prev/next slot
Bank 2 – Trim	P1 start · P2 end (relative to start) · P3 pan · P4 aux send · B1 trigger selected slot · B2 mute selected slot · B3/B4 prev/next
Bank 3 – Filter/Trigger	P1 filter cutoff · P2 filter type · B1 trigger selected slot · B2 apply filter · B3/B4 prev/next

Tip: arm on Bank 1, make a sound to auto-record, navigate slots with B3/B4 (same on every bank), trigger with B1 on any bank. All 8 slots trigger via MIDI notes 36-43 with no cable needed. Each slot also has 5 dedicated CCs (pitch/pan/volume/start/end, from CC50) for bank-independent external automation. Save/load kits over serial with PSSAVE / PSLLOAD, or hold both bank-nav buttons on Bank 1 to browse the 7 most recent kits from the panel.

Piezo Input

CARRIES THE CONTACT MIC SIGNAL

The module that brings Porta's piezo pickup into a patch. Beyond raw audio, it runs trigger detection, envelope following, and pitch tracking on the incoming signal — this is what makes Porta responsive to how something is struck, not just what it sounds like.

INPUTS

gain preamp gain

trig_threshold trigger sensitivity — lower catches softer hits

sensitivity envelope follower tightness

pitch_window pitch detection window — larger = more accurate on low notes, more latency

OUTPUTS

outL mono audio from the mic

trigger gate pulse per detected hit

envelope smooth amplitude follower

pitch detected pitch as CV

Tip: for a responsive contact-mic instrument, patch trigger → ADSR gate, envelope → VCA gain, pitch → Oscillator freq. Tapping objects then produces pitched sound tracking the material's resonance.



Oscillator

Wavetable oscillator with up to 8 drawable waveforms, morphing, and 2-operator FM. Draw custom shapes or use built-in sine/saw/square/triangle presets; morph smoothly between tables with CV or an ADSR for timbral envelopes.

INPUTS

note MIDI note in (pitch+gate+velocity), from MIDIIn/Sequencer

freq raw frequency Hz, for direct pitch control

gain output amplitude 0–1

gate 1=on, 0=off

morph sweep between wavetables 0–1

table_select jump to a wavetable index

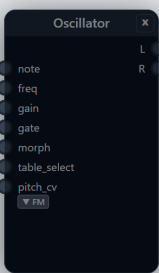
pitch_cv pitch mod in semitones, 0.5=no shift

fm_ratio_cv, fm_amount FM carrier:modulator ratio / depth

fm audio-rate FM in (through-zero FM from another oscillator)

OUTPUTS

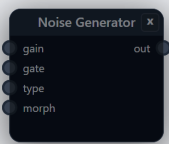
L / R audio out



Tip: basic subtractive voice is Oscillator → Filter → VCA with an ADSR on the VCA gain. Enable FM in the inspector for metallic/bell tones without a second oscillator.

Noise Generator

Multi-type noise generator: white, pink, brown, with a morph control to blend smoothly between them or set a fixed type.



INPUTS

gain output amplitude 0–1

gate on/off, for rhythmic noise hits from a Clock or Sequencer

type 0=white, 1=pink, 2=brown

morph crossfade between types 0–1

OUTPUTS

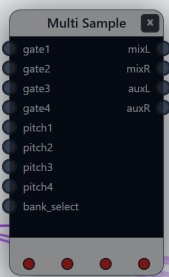
out mono noise

Tip: pink noise into a bandpass Filter makes a wind/ocean texture. Gate with a short ADSR for snare-style hits.

Multi Sample

AUTO-MAPPED · 1 BANK

4-gate multi-sample player, 4 independently-loaded sample slots.



INPUTS

gate1–4 voice triggers

pitch1–4 per-voice pitch

bank_select sample bank

OUTPUTS

mixL / mixR main mix

auxL / auxR aux send

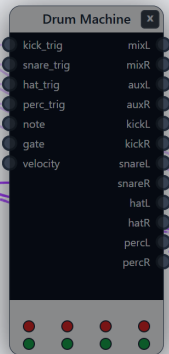
Bank

P1–3 = voice 1 / voice 2 / voices 3–4 levels · P4 = sample bank select (0–3) · B1–4 = trigger voice 1–4

Drum Machine

AUTO-MAPPED · 2 BANKS

4-voice trigger synth — kick, snare, hi-hat, and a percussion voice.



INPUTS

kick/snare/hat/perc_trig per-voice trigger

note, gate, velocity external play

OUTPUTS

mixL/mixR, auxL/auxR main + aux

kick/snare/hat/percL/R per-voice direct outs

Bank 1 — Trigger/Pitch

P1–4 = kick/snare/hat/perc pitch · B1–4 = trigger kick/snare/hat/perc

Bank 2 — Volume/Mute

P1–4 = kick/snare/hat/perc volume · B1–4 = mute kick/snare/hat/perc

Tuned Perc

AUTO-MAPPED • 1 BANK



4-voice tuned percussion using Karplus-Strong physical modelling — marimba, snare, hihat, and tom presets, each with independent pitch, damping, body resonance, and envelope.

INPUTS

gate1–4 trigger voice 1–4

pitch1–4 pitch CV per voice (MIDI note numbers)

OUTPUTS

mixL / mixR stereo mix of all enabled voices

voice1–4_outL/R per-voice direct outs (enable in inspector)

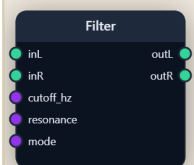
Tip: hold a button + turn its pot for pitch while auto-mapped. Connect a Clock to gates for rhythmic patterns, or a Sequencer to pitch1–4 for melodic Karplus-Strong lines.

Effects

transform an existing signal – need something patched into them

Filter

State-variable filter: lowpass, bandpass, highpass, and notch modes. Shapes the frequency content of any audio signal.



INPUTS

inL / inR audio in

cutoff_hz cutoff freq — modulate with LFO for wah, ADSR for plucks

resonance emphasis at cutoff, 0–1

mode lowpass/bandpass/highpass/notch

OUTPUTS

outL / outR filtered audio

Tip: classic subtractive chain is Oscillator → Filter → VCA. ADSR on cutoff for plucks, LFO on cutoff for slow sweeps/auto-wah.

Delay

AUTO-MAPPED • 1 BANK

Stereo delay with ping-pong mode and freeze (infinite hold). All parameters have CV inputs.

INPUTS

inL / inR audio in

time_ms delay time

feedback 0–1, repeats — above 0.9 risks runaway

mix dry/wet

pingpong 0 = normal stereo, 1 = full L/R bounce

freeze gate: 1 = hold buffer forever

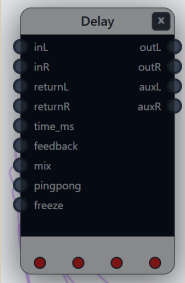
Tip: dub-style echo — feedback 0.6, slow LFO on time_ms. Freeze with a Button for manual capture-and-hold.

Bank

OUTPUTS

outL / outR delayed audio (dry + wet)

P1 time • P2 feedback • P3 pingpong • P4 mix • B1 freeze



Reverb

8-comb Freeverb with 4 serial allpass diffusors — double the comb density of a basic Schroeder reverb, so echoes wash together rather than repeating audibly. Three quality levels trade PSRAM for tail length (Standard ~54KB/37ms, High ~213KB/165ms, Ultra ~687KB/449ms).

INPUTS

inL / inR audio in

mix dry/wet 0–1

size room size — comb feedback gain

damp high-frequency damping

diffuse allpass diffusion — low=grainy/metallic, high=smooth wash

Tip: diffuse 0.7–0.8 for a smooth hall; 0.2–0.3 for a grainier shimmer. Feed into granular with freeze for ambient drones.

OUTPUTS

outL / outR reverb out (dry+wet)



Distortion

Waveshaping distortion with five character presets (Clean, Overdrive, Fuzz, Speaker, Bitcrush) and pre/post tone filtering. CV-selectable preset for rhythmic switching.

INPUTS

inL / inR audio in

drive distortion intensity 0–10

tone post-distortion filter Hz

mix dry/wet

bypass gate: 1=bypass (stomp-box behaviour)

preset_cv CV-selectable preset, 0–1 across presets

OUTPUTS

outL / outR distorted audio



Tip: Overdrive at low drive for warmth; Fuzz/Bitcrush on drums. Drive preset_cv from a Clock to alternate clean/distorted per beat.

VCA

Voltage Controlled Amplifier — the basic building block for shaping dynamics in any patch.

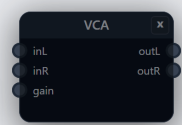
INPUTS

inL / inR audio in

gain amplitude 0–1 — from ADSR for note shaping, LFO for tremolo, or a Pot for manual volume

OUTPUTS

outL / outR audio at current gain



Tip: every voice needs a VCA. Chain two (one from ADSR, one from MIDI velocity) for velocity-sensitive patches.

Pitch Shifter

Time-domain pitch shifter (dual-head ring buffer, Boss PS-style) — shifts pitch up or down in semitones with no external clock needed. Best on monophonic sources: contact mic, voice, single oscillator.



INPUTS

inL mono audio in

semitones CV offset, 0.5=unison, 0–1 maps –12 to +12

fine fine detune CV, ± 100 cents around 0.5 centre

mix dry/wet

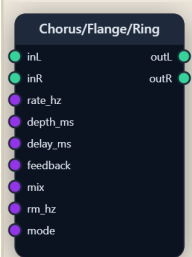
OUTPUTS

outL / outR shifted audio

Tip: fixed intervals via the semitones param — 7=fifth up, 12=octave up. Set `cv_mode: precise` when driving semitones from Scale/ChordGenerator/MIDIIn so it reads as a /48 pitch CV instead of a ± 12 offset.

Chorus / Flange / Ring

Three modulation effects in one, mode-selectable: chorus (subtle doubling), flanger (metallic sweeps), ring modulator (robotic/bell tones).



INPUTS

inL / inR audio in

rate_hz, depth_ms LFO rate/depth for chorus & flanger

delay_ms, feedback base delay & feedback — short delay/high feedback = flanger jet sweeps

mix dry/wet

rm_hz ring mod frequency (ring mode only)

mode chorus / flanger / ring mod

OUTPUTS

outL / outR processed audio

Tip: chorus — rate 0.5Hz, depth 3ms, delay 8ms. Ring mod — try frequencies near musical intervals for metallic harmonics.

Compressor

Dynamics compressor with a stereo-linked envelope follower and sidechain input.

INPUTS

inL / inR audio in

sidechainL / R when connected, drives the envelope follower instead of the main input

threshold level above which compression begins

ratio e.g. 2.0 = 2:1

OUTPUTS

outL / outR compressed audio



Tip: place before Stereo Out to even out dynamics — ratio 2–4, threshold 0.5 as a start. For pumping/ducking, feed a rhythmic source to sidechainL and compress a pad.

Limiter

Hard limiter that prevents audio exceeding the threshold — place at the end of a signal chain to protect against clipping.

INPUTS

inL / inR audio in

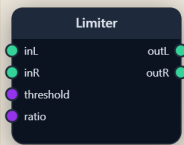
sidechainL / R optional external envelope source

threshold max output level 0–1

ratio higher = closer to brickwall

OUTPUTS

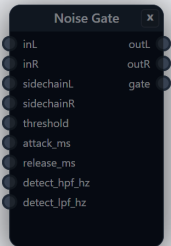
outL / outR limited audio



Tip: place just before Stereo Out as a safety net — threshold 0.8–0.9 catches peaks without audible squashing. Essential after Input Gain or feedback patches.

Noise Gate

Silences audio below a threshold, with a sidechain input and a gate output for triggering other modules. The detector can be band-limited independently of the audio path, turning it into a frequency-selective trigger on any audio-rate source.



INPUTS

inL / inR audio to gate

sidechainL / R optional external level control

threshold level below which audio is silenced

attack_ms, release_ms gate open/close speed

detect_hpf_hz, detect_lpf_hz band-limit the detector only (not the audio) — narrow both to trigger on one frequency range while ignoring others on the same signal

OUTPUTS

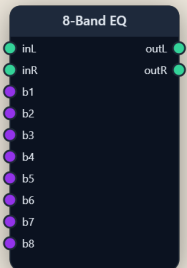
outL / outR gated audio

gate 1=open, 0=closed — trigger envelopes or other events

Tip: a contact mic signal with a low thump and a higher-frequency scrape can trigger on the thump alone by narrowing detect_hpf_hz/detect_lpf_hz around just that range.

8-Band EQ

8-band graphic EQ with fixed frequency bands, each adjustable ± 12 dB.



INPUTS

inL / inR audio in

b1–b8 63/125/250/500/1k/2k/4k/8kHz bands, ± 12 dB each

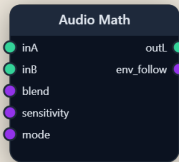
OUTPUTS

outL / outR equalised audio

Tip: cut narrow, boost wide. For contact mic recordings, cut b1–b2 for handling noise, boost b5–b7 for detail.

Audio Math

Multi-mode audio processor: amplitude modulation, crossfade, ring modulation, and vocoder — mode-selectable. Also outputs an envelope follower tracking input A's amplitude.



INPUTS

inA carrier in AM/vocoder; envelope follower tracks this

inB modulator in AM/ring mod, or second crossfade source

blend crossfade position / mod depth

sensitivity envelope follower response speed

mode AM / crossfade / ring mod / vocoder

OUTPUTS

outL processed audio (mono)

env_follow envelope tracking inA, for ducking/triggering

Tip: vocoder mode: inA=voice (modulator), inB=oscillator (carrier). Ring mod with two related-frequency oscillators makes bell tones.

Input Gain

Simple gain stage, 0–4x (12dB), for boosting weak signals like piezo pickups and contact mics.



INPUTS

inL / inR audio in, from LineIn or Piezo Input

gain 0–4x multiplier

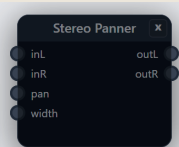
OUTPUTS

outL / outR boosted audio

Tip: follow with a Limiter if boosting aggressively, to prevent clipping downstream.

Stereo Panner

Constant-power stereo panner with width control — positions a signal in the stereo field without a volume dip at centre.



INPUTS

inL / inR audio in (or mono source)

pan 0=left, 0.5=centre, 1=right

width 0=mono, 1=full stereo

OUTPUTS

outL / outR panned audio

Tip: connect an LFO to pan for auto-panning. Set width to 0 to collapse to mono before a mono effect, then re-pan after.

Stereo Mixer (4ch)

AUTO-MAPPED • 2 BANKS

4-channel stereo mixer with per-channel volume, pan, mute, and solo.

INPUTS

1L/1R-4L/4R 4 stereo channel ins

g1-g4 per-channel volume 0-1

master master level

pan1-pan4 per-channel pan

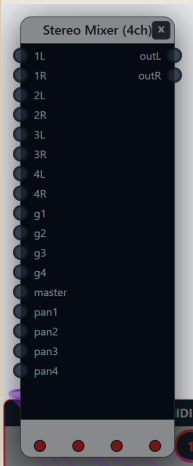
Banks

OUTPUTS

outL / outR summed stereo out

Bank 1: P1-4 = channel volumes, B1-4 = mute • Bank 2: P1-4 = pan, hold encoder + button to solo a channel

Tip: use as the final stage before Stereo Out to mix multiple voices.



Matrix Mixer 2x2

AUTO-MAPPED • 1 BANK

Cross-mix two stereo pairs.

INPUTS

inAL/AR, inBL/BR two stereo pairs in

aa, ab, ba, bb cross-mix amounts

Bank

OUTPUTS

outAL/AR, outBL/BR mixed pairs out

P1-4 = the four routing amounts (A→A, A→B, B→A, B→B), bipolar -1 to +1 each • buttons unused



Modal Resonator

Resonant physical-modelling voice. Ports confirmed from a screenshot; behaviour description pending. Not auto-mapped on Porta — it has fixed hardware mappings on the Eurorack product only, and defaults to no bank (manual patch-to-USB-MIDI-Controls) on Porta.

INPUTS

inL / inR excitation audio

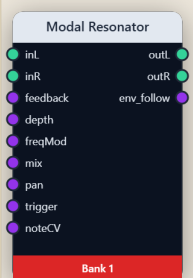
feedback, depth, freqMod resonance shaping

mix, pan, trigger, noteCV mix & pitch control

OUTPUTS

outL / outR resonated audio

env_follow envelope output



Spectral Capture

One-shot FFT spectral analyser. Captures 2048 samples on trigger, runs FFT, extracts the 16 strongest peaks (frequency/amplitude/Q) as a spectral profile that Modal Resonator can load.

INPUTS

inL audio to analyse (piezo, contact mic, any source)

arm rising edge arms capture; next trigger starts it, then auto-disarms

trigger rising edge starts capture (only when armed) — ~46ms at 44.1kHz

threshold noise floor multiplier — higher = fewer, stronger-only peaks

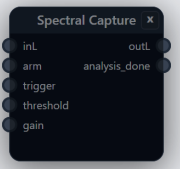
gain input gain before capture

OUTPUTS

outL audio pass-through (with gain)

analysis_done pulses 1.0 for one block when FFT completes — connect to Modal Resonator's trigger to auto-load the profile

Tip: Button/MIDI → arm, Piezo Input trigger → trigger, Piezo Input audio → inL, analysis_done → Modal Resonator trigger. Tap an object and its resonances load into the resonator.



CV Math

Combines two control signals with a selectable operation: add, subtract, multiply, divide, modulo, min, max.

INPUTS

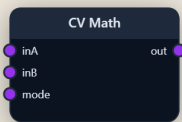
inA, **inB** operands

mode operation-select CV, overrides the mode param

OUTPUTS

out result

Tip: Add mode is the common one — a Pot/CC into A plus an LFO into B modulates around a manually-set centre point. Multiply mode turns it into a VCA for control signals.

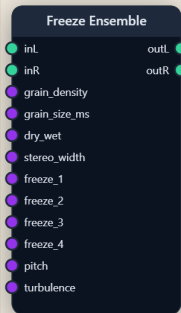


Granular Porta's granular engine (3.4) lives here

Freeze Ensemble

AUTO-MAPPED • 1 BANK

4-layer granular freeze engine. Breaks incoming audio into small grains (roughly 10–100ms) that can be stretched, re-pitched, and re-triggered; four independent freeze layers share a stereo capture buffer. This is the module behind Porta's granular processing (3.4).



INPUTS

inL / inR capture source

grain_density, grain_size_ms grain shape

dry_wet, stereo_width mix

freeze_1–4 per-layer freeze gate

pitch, turbulence pitch + randomisation

OUTPUTS

outL / outR processed audio

Bank

P1 grain_density · P2 grain_size_ms · P3
pitch · P4 dry_wet · B1–4 freeze layers
1–4

Note: stereo_width and turbulence are real parameters but aren't hardware-mapped by default — patch them to USB MIDI Controls if you want them on the panel.

Smudge

AUTO-MAPPED • 2 BANKS

8-grain granular processor, 2-second stereo buffer. Feedback routes grain output back into the buffer — with pitch shift, each pass compounds into shimmer cascades. Sub-30ms grains at high density create metallic/formant textures.

INPUTS

inL / inR audio to fill the grain buffer

grain_density grains/sec, 1–80

grain_size_ms 5–500ms — under 30 = metallic, 200+ = pad-like

position, position_spread buffer read point + per-grain randomisation

pitch, pitch_spread, pitch_interval pitch shift + randomisation + a second alternating pitch (7=fifth, 12=octave)

dry_wet, feedback mix + grain-output-into-buffer amount

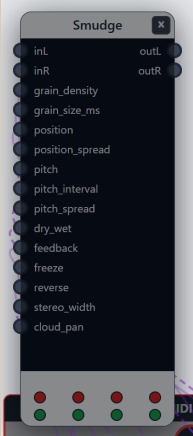
freeze gate: 1 = stop recording, replay frozen buffer

reverse probability of a grain playing backward

stereo_width, cloud_pan stereo spread + overall cloud position

OUTPUTS

outL / outR granular audio



Banks

Bank 1: density/size/position/feedback + freeze/reverse • Bank 2: pitch/stereo_width/pitch_spread/dry_wet

Tip: shimmer reverb — pitch=+12, feedback=0.6, position=0.1, high density. Granular delay — pitch=0, feedback=0.7, position near 0.

Spectral Gate

Two-input spectral gate with granular capabilities. Input A (source) is optionally band-filtered, then gated/scattered by Input B (exciter) — at scatter=0 it's a clean sidechain gate, at scatter=1 exciter transients trigger grains from a 2-second source buffer. Band filter isolates a frequency range; only that band is affected, everything else passes clean.

INPUTS

1L/1R source (sample/recording)

2L/2R exciter (sidechain, e.g. contact mic, drum loop)

sensitivity onset/gate threshold

scatter 0=clean gate, 1=full granular

band_freq, **band_q** target band centre
(0=fullband, CV 20Hz–10kHz) + width

grain_density, **grain_size_ms** grain rate
(0=trigger-only) + duration

position, **position_spread**, **pitch**,
pitch_spread grain read/pitch controls

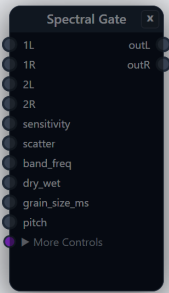
feedback, **dry_wet**, **exciter_mix**,
freeze mix controls

stereo_width, **cloud_pan**, **trigger** stereo
spread, pan, external trigger CV

Tip: sidechain gate — scatter=0, band_freq=0, drum into exciter. Granular transplant — scatter=1, density=0, contact mic as exciter.

OUTPUTS

outL / outR processed audio



Sampling / Looping record, loop, and slice audio



Looper

AUTO-MAPPED · 1 BANK

20-second PSRAM looper with clean additive overdub, adjustable feedback, reverse playback, variable tape speed, and non-destructive loop-region selection. Dry signal always passes through.

INPUTS

inL / inR audio to record

record gate: 1=recording/overdubbing, 0=playback

decay overdub feedback — 1.0=infinite clean overdub, 0.707=Frippertronics fade, 0=replace

reverse short pulse toggles reverse, hold (>500ms) reverses while held

speed playback speed, 0.5=normal, exponential 0.25x–4x

loopStart, loopSize loop region position/size 0–1, for scrubbing/isolating parts of the buffer

OUTPUTS

outL / outR audio (dry + loop)

position playback position 0–1

Tip: decay=1.0 layers overdubs infinitely with zero quality loss; 0.7 gives Frippertronics-style build-then-fade. Modulate speed with an LFO for tape-warble.

BPM Looper

AUTO-MAPPED · 2 BANKS

4-slot beat-synced looper with slice-based randomisation and per-slot pitch control. Each slot records a different beat length.

INPUTS

inL / inR audio to record

clock tempo sync

vol1-4, rec1-4 per-slot level/record

rand1-4, pitch1-4 per-slot randomise/pitch

OUTPUTS

outL / outR mixed looper output



Bank 1 – Record/Volume

P1-4 = slot 1-4 volume · B1-4 = per-slot record/play/mute (tap cycles state, hold 1.5s clears)

Bank 2 – Randomise/Pitch

P1-4 = slot 1-4 slice-randomisation amount · buttons unused (pitch is CV-only, no panel control)

Tip: record different length loops (1/4 beat, 1 beat, 4 beats) for polyrhythmic layering; connect a Clock to keep everything synced.

Paralloop

AUTO-MAPPED · 2 BANKS

8-layer parallel looper — each overdub goes to its own buffer, so layers never degrade. First recording sets the master loop length (max 1.5s); later layers auto-record for the same duration. Envelope followers generate 4 trigger-bus outputs that can drive Drum Machine or Multi Sample. ~3MB PSRAM.

INPUTS

inL / inR audio in

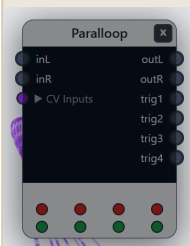
ploop_speed per-layer speed CV, 0.5=no change

ploop_mix dry/wet

OUTPUTS

outL / outR mixed output

trig1-4 trigger bus outs from layer envelope followers



Bank 1

B1 record/play · B2 mute · B3 lock · B4 reverse · P1 speed · P2 level · P3 loop start · P4 loop length

Bank 2

B1/B2 navigate layers · B3 mixdown (merge unlocked layers) · B4 undo · P1-4 trigger routing (0=none, 1-4=bus) for 4 layers at a time

Tip: lock a layer to protect it from mixdown. Mixdown merges unlocked layers to free slots for more recording.

Audio Slicer

AUTO-MAPPED · 3 BANKS

Note-triggered slice playback.

INPUTS

inMono, inL, inR source audio

note trigger

slice_count, pitch_cv slicing + pitch

OUTPUTS

outL / outR playback

auxSendL/R aux

slice, slicePosition state out

Bank 1 – Main

P1 = pitch (± 1 octave, centre detent) ·
P2 = volume (centre = unity) · P3-4 =
start/end (reserved) · B1 = record/play
(short) or clear/load (long) · B2 =
forward/reverse (short) or save to SD
(long) · B3/4 = previous/next slice

Bank 2 – Slice ADSR

P1-4 = attack/decay/sustain/release ·
same B1-4 as Bank 1

Bank 3 – Aux Send

P1 = aux send level, P2-4 unused · same
B1-4 as Bank 1



Gene Looper

AUTO-MAPPED · 1 BANK

Generative/mutating looper — each recorded slot can be scrubbed through algorithmic variations ("genes") rather than just looping verbatim.

INPUTS

inL / inR audio to record

record, reverse, slot_select, pan
transport

genes, gene1-4 mutation control

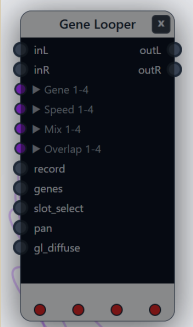
speed1-4, mix1-4, overlap1-4 per-slot
playback

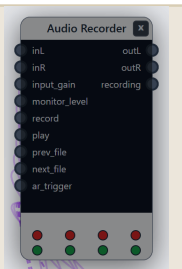
OUTPUTS

outL / outR looper output

Bank

P1 = gene selection (scrubs through
divisions) · P2 = speed/pitch (centre =
unity) · P3 = diffuse (per-slot allpass
blur) · P4 = pan · B1 = start/stop
recording the current slot · B2 = cycle
gene count (2/4/8/16/32/64) · B3 =
reverse toggle · B4 = advance to the
next of 4 recording slots





Audio Recorder

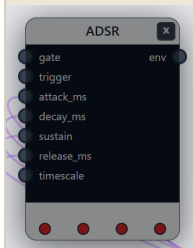
AUTO-MAPPED · 1 BANK

Records to SD as WAV with auto-incrementing filenames. Also handles playback of files already on the card, not just capturing new ones (4.1).

Bank

B1 record · B2 play · B3 previous file ·
B4 next file

Control generate signals that move other modules' parameters



ADSR

AUTO-MAPPED · 1 BANK

Envelope generator shaping a signal over time using Attack, Decay, Sustain, Release stages. Also supports freeform breakpoint mode for complex shapes.

INPUTS

gate 1 = start attack, 0 = start release

trigger re-triggers without waiting for release
— drum hits

attack/decay/release_ms, sustain stage
modulation

timescale scales all times

OUTPUTS

env envelope output, 0–1

Tip: gate from a Button or MIDI In, route env to a VCA's gain. Loop/pingpong mode gives self-running LFO-like shapes.

Bank

P1 attack_ms · P2 decay_ms · P3 sustain
· P4 release_ms



LFO

Low Frequency Oscillator for modulation. Produces slow, repeating waveforms to add movement to any parameter — vibrato, tremolo, filter sweeps, panning.

INPUTS

rate_hz speed — patch a Pot for manual control

wave sine/saw/square/triangle

depth, offset amplitude & bias

clock tempo sync

OUTPUTS

out modulation signal

Tip: to Oscillator pitch_cv for vibrato, to Filter cutoff for wah. Clock-sync locks the rate to your patch tempo.

Clocks

AUTO-MAPPED · 1-2 BANKS

Master clock/divider, up to 8 divided pulse outputs.

INPUTS

bpm_input, **rate1-4** tempo control

start_stop transport

OUTPUTS

out1-8 divided pulses

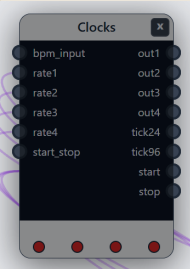
master_phase, **master_pulse**, **tick24**,
tick96 sync outs (+2 more not yet catalogued)

Bank 1

P1-4 = output 1-4 clock division ($\div 16$ to $\times 16$) · B1 = start/stop · B2 = tap tempo
· B3 = reset to beat 1 · B4 unused

Bank 2 (if use_outputs_5_8 is on)

P1 = divisions for outputs 5-8, rest unused



Step Sequencer

8-part, 16-step trigger and velocity sequencer with a built-in clock. Each step cycles off/low/med/high velocity on repeated press; per-part loop start/end points allow polyrhythmic patterns. Click the title to open the grid editor.

INPUTS

clock external clock or per-hit trigger — wire a piezo/drum pad here to step gesturally, one hit = one step (bypasses transport, disables PLAY/STOP while connected)

reset resets all parts to step 0

start_stop transport 1=run, 0=stop

bpm BPM CV, overrides internal BPM param

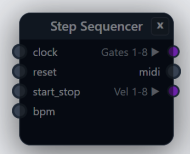
OUTPUTS

gate1-8 per-part gate (high when step active)

vel1-8 per-part velocity (0/0.33/0.66/1.0)

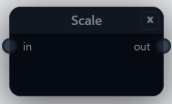
midi combined MIDI out for all 8 parts (notes 36-43 default) via a MIDIOut module

Tip: gate outs → Drum Machine triggers, velocity outs → velocity ins. Double-tap a part button in the editor to set loop start/end for polyrhythms (e.g. 16 vs 12 steps = 4:3).



Scale

Maps or quantises a control signal to a musical scale. **select** mode maps a 0–1 CV across N scale notes (a pot/LFO picks which note plays); **quantise** mode snaps an incoming pitch CV to the nearest in-scale note.



INPUTS

in **select** mode: 0–1 CV (Pot, LFO). **Quantise** mode: /48-encoded pitch CV (MIDIIn, Sequencer, Piezo Input)

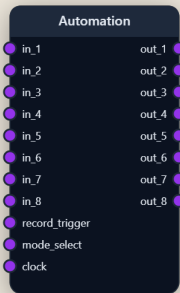
OUTPUTS

out pitch CV, /48 encoding (0.5 = MIDI 60) — connect to an Oscillator's note input

Tip: for a noisy pitch source like a contact mic on a tongue drum, set mode=quantise and pick a scale matching the drum's tuning — the oscillator then always plays in-key regardless of pitch-tracking error.

Automation

Records and loops control movements on up to 8 channels. Long-press the encoder to record pot/button automation, then play it back in a loop. Inputs expand dynamically as you connect more.



INPUTS

in_1–8 channel inputs — connect any control source to record

record_trigger start/stop recording, from encoder or external gate

mode_select record vs playback

clock external clock for tempo-synced playback

OUTPUTS

out_1–8 channel playback

Tip: connect a Pot to in_1, long-press the encoder, sweep the pot, stop — out_1 replays that movement in a loop.

Harmonic Sequencer

AUTO-MAPPED • 2 BANKS

Multi-step pitch/gate/velocity sequencer.

INPUTS

clock, **start_stop**, **gate** transport timing
swing, **gate_length**, **bpm**, **clock_div**

OUTPUTS

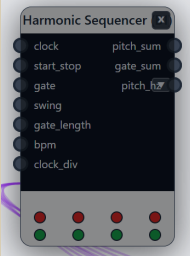
note, **pitch/gate/velocity_sum** summed outs
pitch_hz + **per-step outs** (+13 more not yet catalogued)

Bank 1

P1-4 = step 1-4 pitch (2-octave range each) · B1-4 = toggle mute for step 1-4

Bank 2

Reserved by the module (claims 2 banks total) but no pot/button mapping is documented in the module registry – likely unused or still to be wired up.



Gate Generator

Generates rhythmic gate pulses from a clock (internal or external), with a built-in Sample & Hold and a probability control for rhythmic randomness.

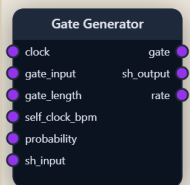
INPUTS

clock external clock, each pulse triggers a gate
gate_input manual gate control, overrides clock triggering
gate_length fraction of clock period the gate stays high
self_clock_bpm internal-clock BPM when nothing's connected to clock
probability chance of firing per tick, 0–1
sh_input signal sampled on each clock tick

OUTPUTS

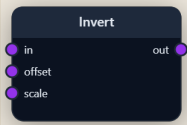
gate output, high on tick (subject to probability)
sh_output Sample & Hold value, held steady between ticks
rate current clock rate as CV, for syncing other modules

Tip: connect an LFO to sh_input and use sh_output for stepped random melodies. Probability below 1.0 thins out a drum pattern for natural variation.



Invert

Inverts a control signal (1-x), with optional scale and offset — turns high values low and low values high.



INPUTS

in signal to invert, 0–1

offset added after inversion, shifts the range

scale multiplies the inverted signal

OUTPUTS

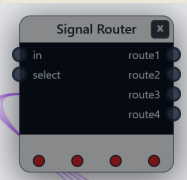
out inverted signal

Tip: place between an ADSR and a Filter cutoff for a ducking effect — the filter closes as the envelope opens. Also gives opposite modulation on two parameters from one LFO.

Signal Router

AUTO-MAPPED · 1 BANK

Routes one input signal to one of 4 outputs (all others silenced) — works with audio or control signals. Useful for directing a trigger or audio signal to different destinations without re-patching.



INPUTS

in signal to route (audio or control)

select CV select: 0.0/0.33/0.67/1.0 → route1–4, overrides the select param

OUTPUTS

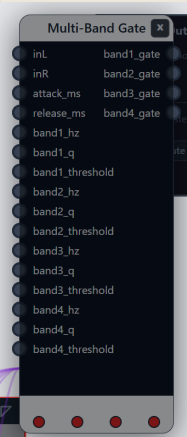
route1–4 active when select=0/1/2/3 respectively

Tip: Piezo Input trigger → in, route1–4 → Drum Machine's kick/snare/hat/perc_trig. Press Btn1–4 to choose which drum to hit.

Multi-Band Gate

AUTO-MAPPED · 1 BANK

Splits one audio signal into 4 independent trigger/gate outputs, each responding only to its own frequency band via a state-variable band-pass filter feeding its own envelope follower and threshold gate. When auto-mapped, Btn1–4 select which band Pot1–3 edit (threshold/Q/cutoff); each band remembers its own settings when switching away and back. The 4 button LEDs double as a live gate-activity monitor for all 4 bands simultaneously.



INPUTS

inL / inR audio to analyze (summed to mono internally — all 4 bands share one detector)

attack_ms shared envelope attack across all 4 bands

OUTPUTS

band1–4_gate per-band trigger/gate out

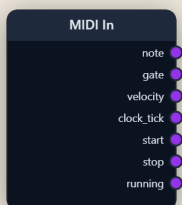
Bank

B1–4 select band · P1 threshold · P2 Q · P3 cutoff Hz

Hardware I/O where a patch touches Porta's physical world

USB MIDI In

USB MIDI input only — no TRS. Porta appears as a class-compliant USB MIDI device to a computer, with roughly 1–3ms latency. Unrelated to the physical MIDI TRS jacks (item 5, 1.4).



INPUTS

— no inputs

OUTPUTS

note MIDI note as 0–1 CV

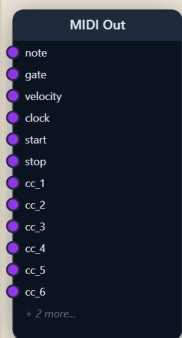
gate note on/off

velocity 0–1

Tip: use instead of MIDI In when you only need USB MIDI — connect from a DAW to play the synth from a computer keyboard or sequencer.

USB MIDI Out

USB MIDI output with up to 8 groups, each routable to a different MIDI channel — sends notes/CCs to a computer or DAW over USB. Ports expand dynamically as you connect more, same behaviour as Automation. USB-only, like USB MIDI In — unrelated to the hardware jack (item 6, 1.4).



INPUTS

note_1-8, gate_1-8, velocity_1-8, cc_1-8 per-group note/gate/velocity/CC in

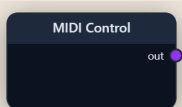
OUTPUTS

— no control outputs

Tip: route different sequencer outputs to separate groups with different channels, to control multiple DAW instruments from one patch.

MIDI Control

Virtual MIDI CC knob or note gate, reading from USB or hardware MIDI. One instance = one CC (or note gate), hence the single out port and no inputs. This is the building block behind the USB MIDI Controls strip (3.2, FIG 3.2a) — a bank's four pot slots there are four MIDI Control instances the patcher creates and wires automatically, not a separate bidirectional module. Add extra instances manually to build a larger virtual control surface than the strip's four slots.





Stereo Out

Stereo audio output to the board DAC — every patch needs at least one to produce sound. On Porta, this is the module feeding the exciter jack (item 9, 1.4); there's no separate "Exciter Out" module.

INPUTS

L left channel — connect from any audio output

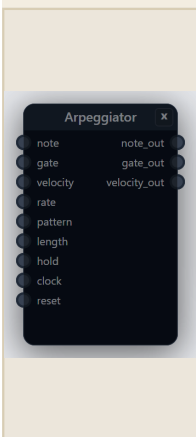
R right channel

OUTPUTS

—

Tip: for mono, connect the same source to both L and R. For stereo, use a Panner or Stereo Mixer before this module.

MIDI Effects process note data, not audio



Arpeggiator

Note pattern generator. Ports confirmed from a screenshot; behaviour description pending. Not auto-mapped — patch its parameters to USB MIDI Controls to get them on the panel.

INPUTS

note, gate, velocity held chord in control

rate, pattern, length, hold pattern control

clock, reset sync

OUTPUTS

note_out, gate_out, velocity_out arpeggiated output



MIDI Looper AUTO-MAPPED · 1 BANK

Records/plays back MIDI note data.

INPUTS

note, gate, velocity notes to record

clock sync

Bank

OUTPUTS

note/gate/velocity_out playback

playing, recording, beat state

P1 = playback velocity · P2 = transpose (±24 semitones) · P3 = timing randomization · P4 = pitch randomization · B1 = record/play state machine (arm → count-in → record → play → overdub → stopped) · B2 = hold for scale-quantize lock · B3 = long-press to clear · B4 unused

The Patcher interface, at a glance

Everything about the patcher itself referenced elsewhere in this manual, gathered in one place:

Canvas	Where modules are placed and cabled together. Dulled/dimmed with nothing connected; brightens once Porta is paired.
Cable colours	Green = audio. Purple = data/control.
USB MIDI Controls	The module that bridges a patched parameter to the physical panel (3.2).

Toolbar (desktop)

Real screen capture, left to right as they appear once connected:

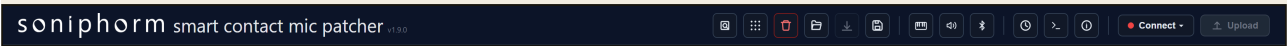


FIG A.5a — Toolbar, desktop width.

-  **Fit** (Zoom to Fit) — reframes the canvas view to show every module on the current patch at once.
-  **Tidy** (Snap to Grid) — aligns every module to a grid for a cleaner layout.
-  **Clear** (Clear Canvas) — deletes every module and cable on the canvas. No confirmation step, so it's worth a moment's pause before clicking.
-  **Open** (Load Patch from File) — loads a previously saved patch file from disk.
-  **Recall** (Download Patch) — pulls whatever patch is currently loaded on the connected Porta back into the canvas, for picking up where a previous session left off. Greyed out until a device is connected.
-  **Save** (Save Patch to File) — saves the current canvas as a patch file to disk.
-  **Keys** (Show/hide keyboard) — toggles an on-screen musical keyboard for playing or testing a patch without external MIDI gear connected.
-  **Mute** — silences Porta's audio output without disconnecting or stopping the patch.
-  **BLE Ctrl** (desktop/USB only) — hands control to the separate BLE MIDI Controller app, opened in a new tab, which streams a connected phone/controller's pot and button values to Porta over Bluetooth. Only shown when connected over USB — when connected over BLE already, the patcher's own perform strip is the controller and this button doesn't apply (see Mobile, below).
-  **History** (Upload History) — lists previously uploaded patches from this browser, recallable back onto the canvas.
-  **Console** — opens the serial console: raw device log output and a command line for the serial commands in the Reference appendix.
-  **Info** (Device Information) — shows connection and firmware version status; surfaces the firmware update flow when a newer version is available (6.3). A small amber dot on the icon means an update is ready.
-  **Connect** — opens a dropdown to choose USB Serial, Bluetooth, or WiFi (if enabled) as the transport. The coloured dot shows live status: red = disconnected, green = connected over USB or BLE, purple = connected over WiFi.
-  **Upload**

Upload — always visible regardless of screen size. Pushes the current canvas to Porta as a patch, prompting for a name and hardware slot first (3.2). Disabled until a device is connected.

Mobile differences

Below 768px width, the toolbar and the module index/inspector columns switch from always-visible to on-demand, to fit a phone or tablet screen:



FIG A.5b — Mobile toolbar, closed. Only the hamburger menu and Upload remain visible; everything else collapses behind the menu.

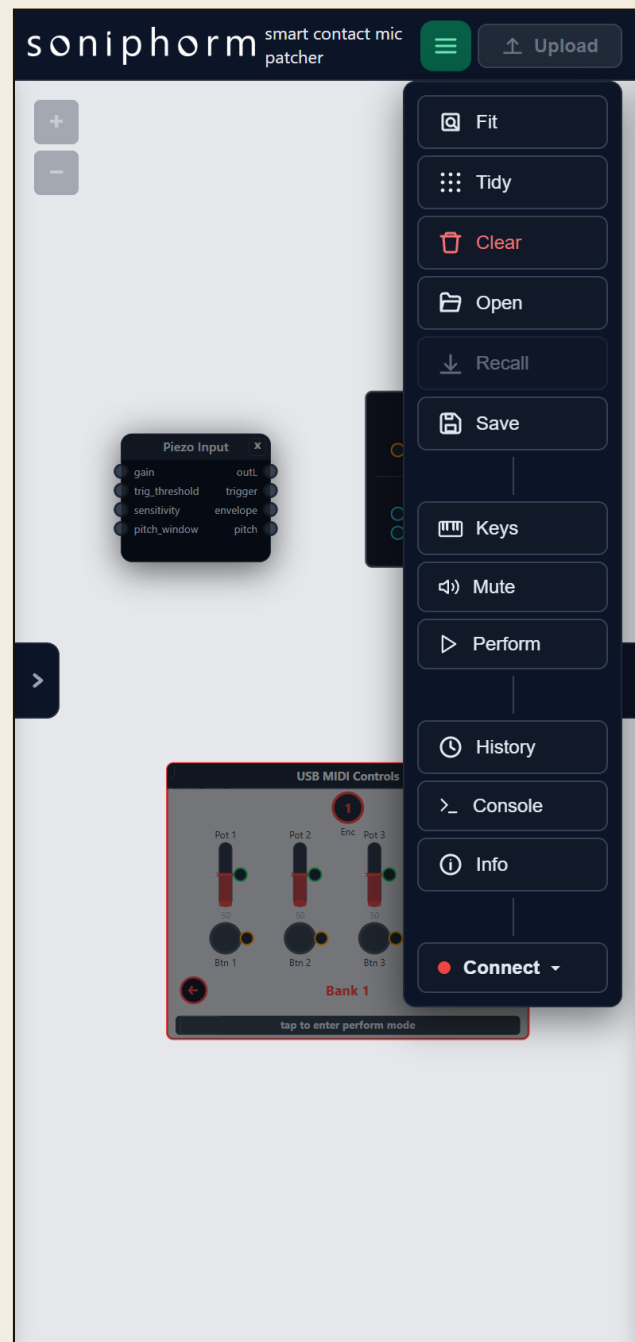


FIG A.5c — Hamburger menu open: the same tools as the desktop toolbar, now stacked with labels instead of an icon-only row.

Toolbar

Collapses to a hamburger menu (≡) plus Upload. Tapping the hamburger opens every desktop tool listed above as a labelled vertical list instead of a horizontal icon row.

Module index & inspector

Off-screen by default rather than fixed side columns, to leave the canvas full width. A small arrow tab on the left screen edge opens the module index; a matching tab on the right opens the inspector – tap either to slide that column in over the canvas. With the column open, its tab now points the other way; tapping it again slides the column back out. Left tab and left column, right tab and right column, always paired the same way.

Perform

A Perform button (play-triangle icon) appears only on mobile, hidden on desktop. Tapping it dims the cables and turns the canvas itself into a touch control surface – mobile's equivalent of the desktop's separate BLE MIDI Controller app.

BLE Ctrl

Hidden entirely on mobile, for the same reason: there's no separate controller app to hand off to, because Perform mode on the phone/tablet screen already is the controller.

TIP

For a walkthrough of the canvas itself — placing a module, drawing a cable — see the video tutorials at youtube.com/@soniphorm.